Epidemiology: The Science of Public Health

From Core Principles to Outbreak Response



The Backbone of Public Health

“Epidemiology is the backbone of public health.”

It is the basic science of public health, essential for identifying and mapping emerging diseases. It not only helps track diseases but also informs strategies for prevention and control, playing a pivotal role in combating recent global health threats like COVID-19.



Individuals

Observation of effects on individuals (e.g., patient history).

Laboratory

Laboratory experiments (e.g., drug efficacy trials).

Populations

Measuring their distribution in a population (epidemiology).

What is Epidemiology?

Epidemiology is the **study** of the **distribution** and **determinants** of health-related states or events in **specified populations**, and the **application** of this study to the prevention and control of health problems.

Source: From the Greek ‘epi’ (upon), ‘demos’ (people), and ‘logos’ (doctrine).
Translation: “The doctrine of what is upon the people.”

The Five Core Components



1. Study

Includes surveillance, observation, hypothesis testing, analytic research, and experiments.



2. Distribution

Refers to the analysis of times, persons, places, and classes of people affected.

Epidemiology



4. Health-Related States & Events

Refers to diseases, causes of death, behaviours (like tobacco use), positive health states, and the provision/use of health services.



3. Determinants

Includes factors that influence health: biological, chemical, physical, social, cultural, economic, genetic, and behavioural.



5. Application

The aims of public health—to promote, protect, and restore health.

The Epidemiological Framework: A Series of Questions

The key components of data needed for health planning, supervision, and evaluation can be approached through a series of questions.



What?

What is the disease or condition? Its clinical manifestation and diagnosis.



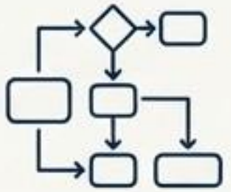
Who?

Who is affected? Referring to age, sex, social class, occupation, etc. (Person factors).



Where?

Where did it happen? In relation to place of residence, geographical distribution, and place of exposure. (Place factors).



How?

How did the disease occur? The interplay of agent, vector, source, and susceptible groups.



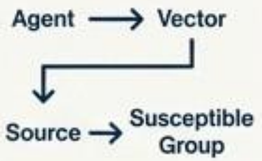
Why?

Why did it occur? The reasons for the disease outbreak.



When?

When did it happen? In terms of months, season, or year. (Time factors).



How?

How did the disease occur? The interplay of agent, vector, source, and susceptible groups.



What Now?

The most important question. What action is to be taken as a result of the information gained?

The Concept of Multiple Causation: The Epidemiological Triad

Health and disease are outcomes of the dynamic relationship between an agent, a host, and the environment. A state of equilibrium indicates no disease; any disturbance results in disease.



Agent: The factor whose presence or absence causes disease.

- Biological (viruses, bacteria)
- Chemical (lead, solvents)
- Physical (heat, radiation)
- Nutritional (excess or deficiency)
- Mechanical (bullets, knives)
- Social/Psychological (poverty, work stress)

Host: The person or animal that harbours the disease.

Factors influencing exposure/response:

- Age, sex, occupation
- Customs
- Inherent/acquired immunity

Environment: External conditions influencing the interaction.

Examples:

- Biological (vectors, reservoirs)
- Social (economic/political organisation)
- Physical (heat, light, air, water)

Evolving Theories of Disease Causation



Germ Theory (Monocausal)

The agent must always be found with the disease in question and not with any other disease. Could not explain why not all exposed individuals become ill.



Epidemiological Triad

Sees disease as the product of an interaction between an agent, a host, and the environment. Useful for infectious disorders, but less so for chronic diseases.



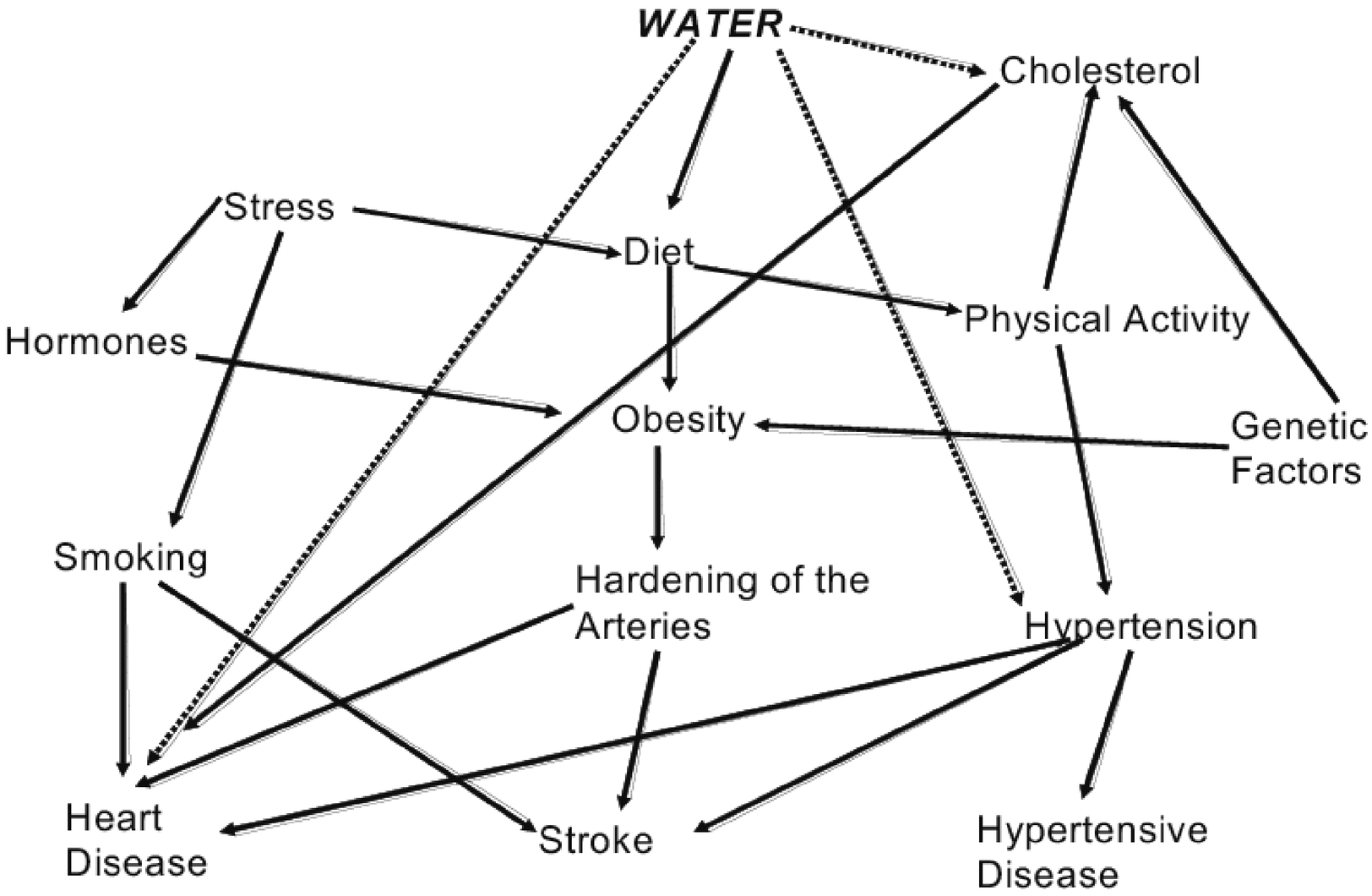
Web of Causation

Considers all predisposing factors and their complex interrelationships. The disease is an outcome of the interaction of multiple factors. Ideally suited for the study of chronic diseases where the agent is often not known.



Socio-Environmental Approach

Seeks to understand why some social groups are more susceptible to disease in general. It focuses on the population and identifies factors that make and keep people healthy, forming the basis for health promotion strategies.


WATER
Cholesterol
Stress
Diet
Physical Activity
Hormones
Obesity
Genetic Factors
Smoking
Hardening of the Arteries
Hypertension
Heart Disease
Stroke
Hypertensive Disease

Calibrating the Threat: Understanding Levels of Disease

The amount of a disease usually present in a community is the **baseline** or **endemic level**. This is the observed, expected level.

Sporadic



A disease that occurs infrequently and irregularly.

Endemic



The constant presence and/or usual prevalence of a disease within a geographic area. (e.g., malaria in parts of Africa).

Epidemic / Outbreak

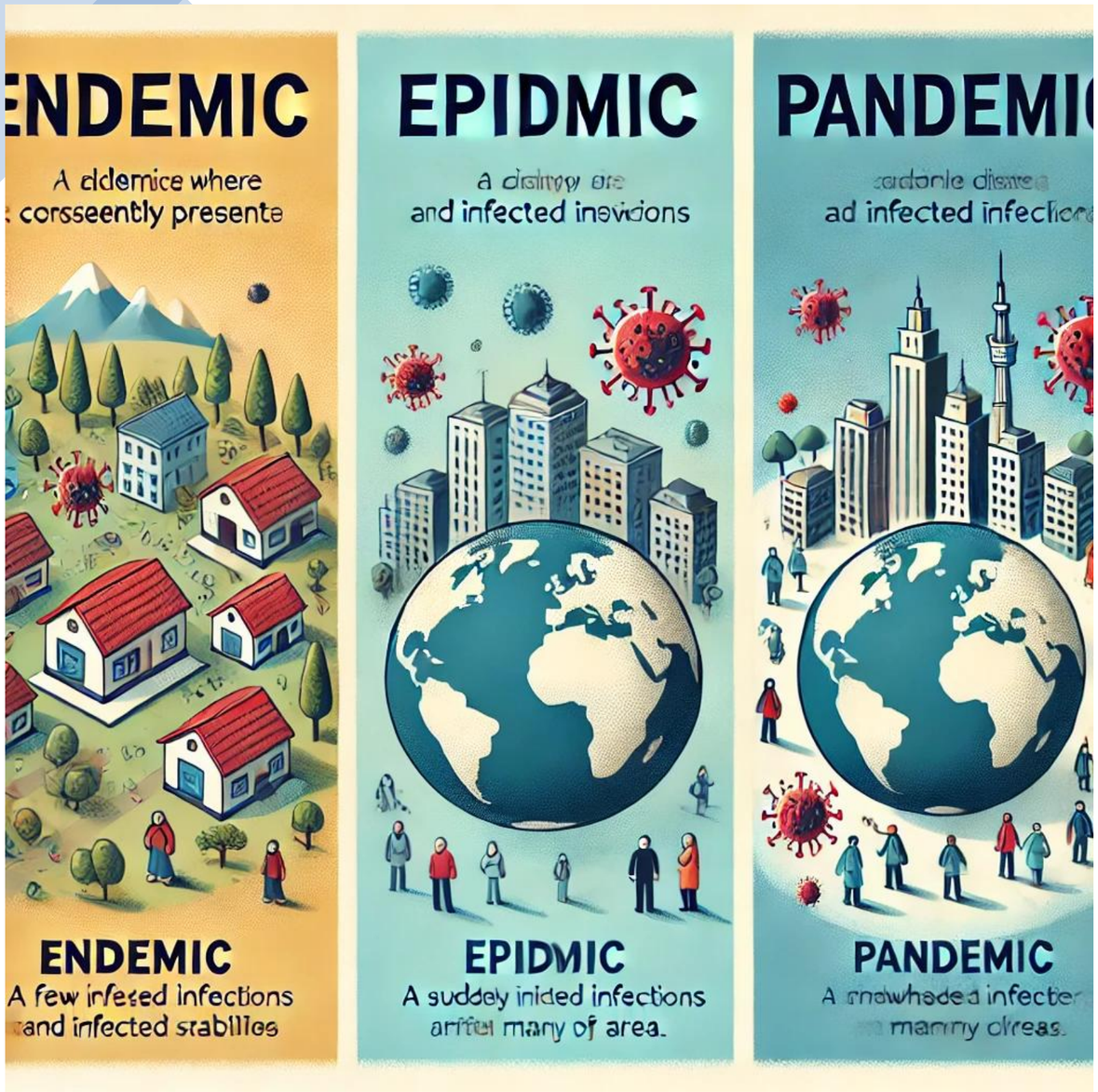


An increase, often sudden, in the number of cases of a disease above what is normally expected in that population in that area. (e.g., H1N1 flu).

Pandemic



An epidemic that has spread over several countries or continents, usually affecting a large number of people. (e.g., COVID-19).


ENDEMIC
A few infected infections
and infected stabilities
EPIDMIC
A suddely inided infections
PANDEMIC


BROAD STREET
PUMP
GREAT PULTENEY STREET
HUSBAND ST
PETER

Identifying an Outbreak: When the Numbers Rise

An outbreak is the occurrence of a disease or any health-related event in a community in excess of the expected level.

What is "excess"? It depends on the disease and the community.

One case of Polio in Palestine?

YES, an outbreak.

The expected number is zero.

Five cases of Rabies in a city (where 2 is normal)?

YES, an outbreak.

Even though the number is small, the disease causes death.

110 cases of Cutaneous Leishmaniasis in a city (where 100 is normal)?

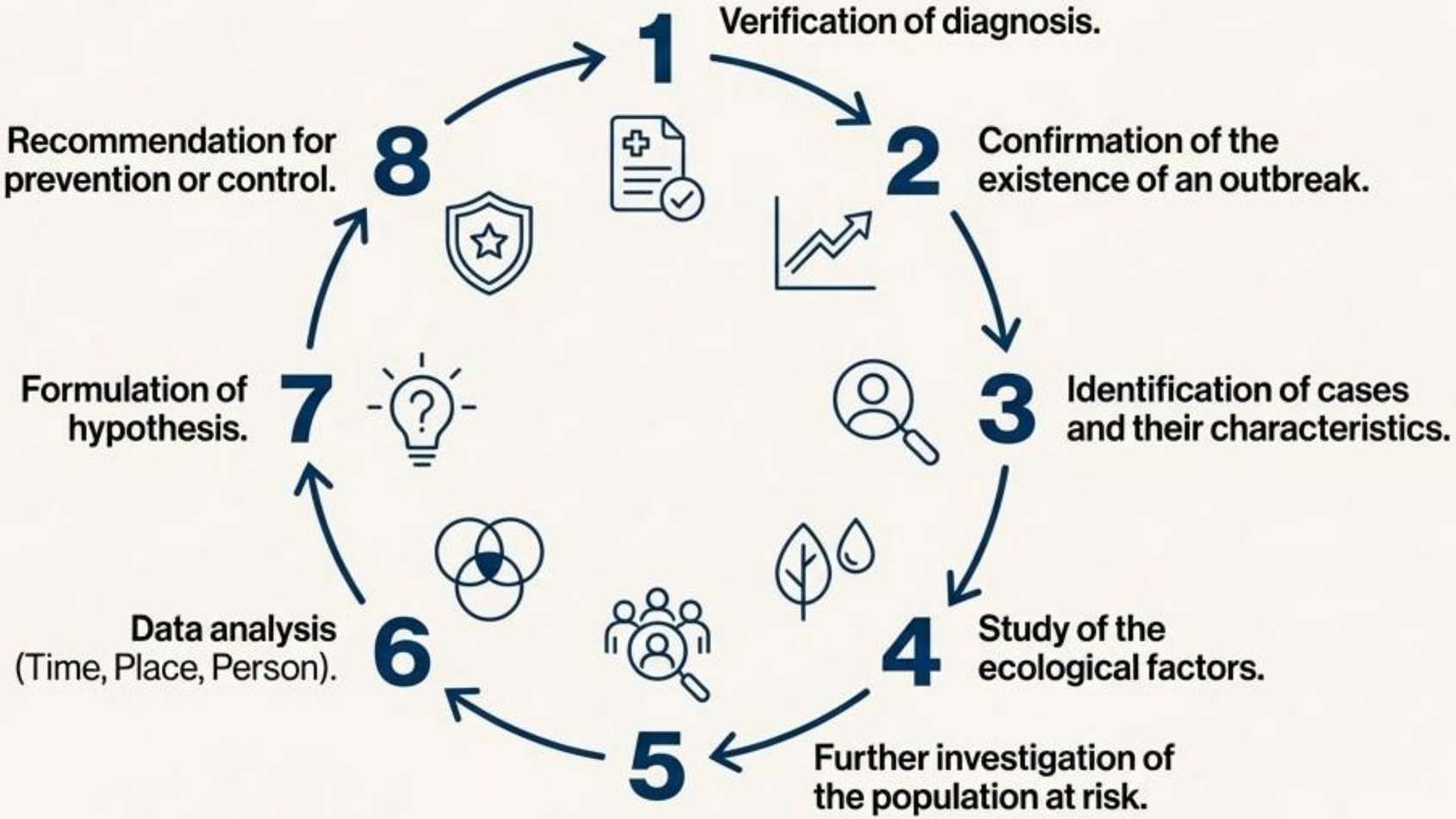
MAYBE NOT.

The disease is not very serious. A big rise (e.g., 150 cases) would definitely be an outbreak.

**The terms 'outbreak' and 'epidemic' have the same meaning. 'Outbreak' is often used for a more limited geographic area or to prevent public fear.*

The Response: A Systematic 8-Step Investigation

Investigating an outbreak is a duty of public health professionals. It requires a systematic and scientific approach to find the cause, control the spread, and prevent morbidity and mortality.



The Investigation Protocol, Steps 1-4: Establishing the Facts

1 Verify Diagnosis

Confirm the initial reports are diagnostically correct. (e.g., Is it really Smallpox, or is it Chickenpox?).



2 Confirm Outbreak

Check records to ensure the number of cases is truly above the expected incidence for the area.



3 Identify Cases & Characteristics

Find all cases (from hospitals, labs) and gather information about them, including exposure to risk factors (e.g., source of food/water in a typhoid outbreak).



4 Study Ecological Factors

Investigate environmental factors that may have made the outbreak possible (e.g., change in temperature, movement of people, increase in mosquito vectors for a malaria outbreak).



The Investigation Protocol, Steps 5-8: From Analysis to Action

5 Investigate Population at Risk

Obtain additional information from the population, if necessary (e.g., immune status, blood examinations).



6 Data Analysis

Describe the cases in terms of Time, Place, and Person. This descriptive epidemiology helps understand the outbreak in detail.



7 Formulate Hypothesis

Based on the descriptive data, develop a hypothesis about the cause and spread of the outbreak.



8 Recommend Prevention & Control

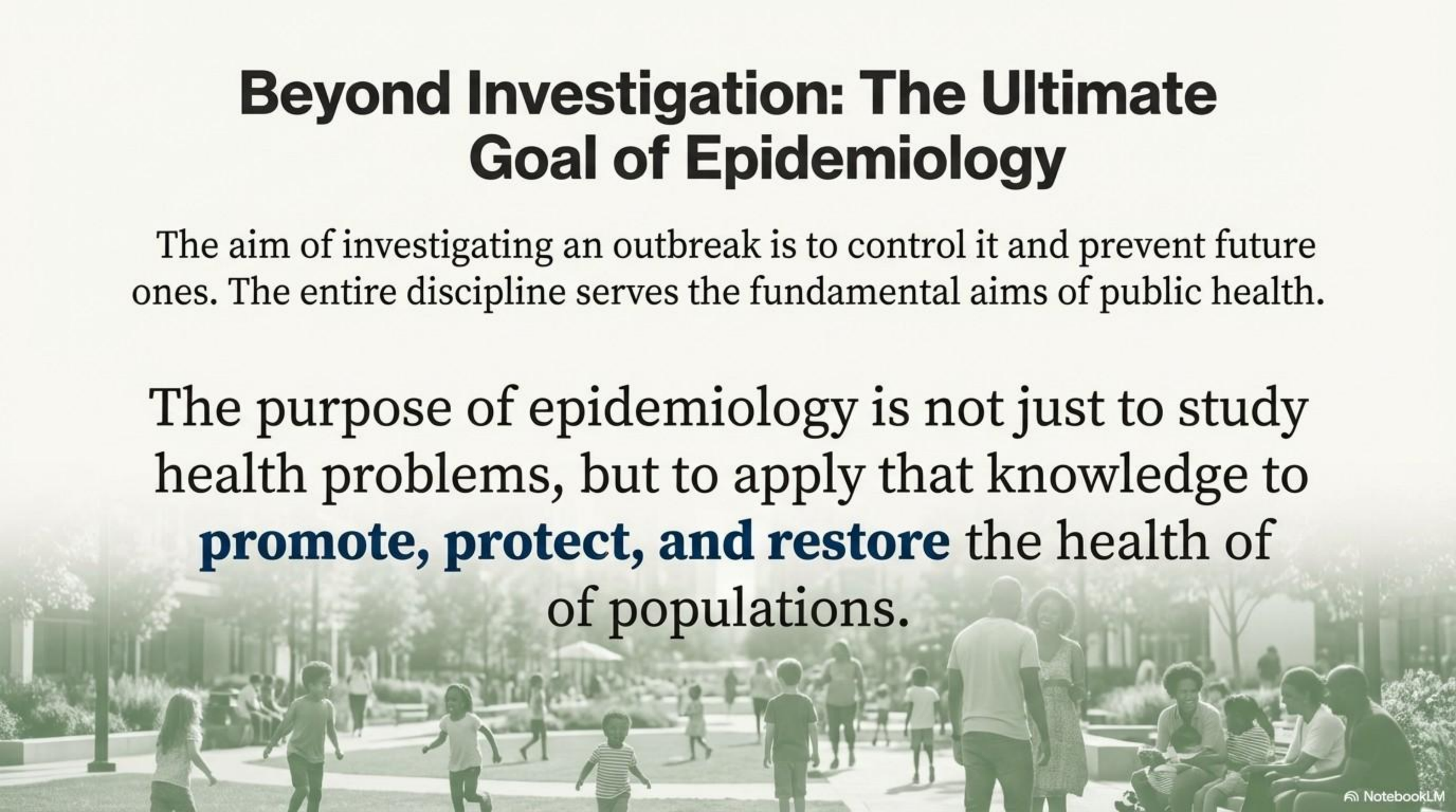
Based on the findings, make recommendations to control the current outbreak and prevent future ones.



Beyond Investigation: The Ultimate Goal of Epidemiology

The aim of investigating an outbreak is to control it and prevent future ones. The entire discipline serves the fundamental aims of public health.

The purpose of epidemiology is not just to study health problems, but to apply that knowledge to **promote, protect, and restore** the health of of populations.



NotebookLM

Thank You

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